Requirements Specification

Pharmacy Expiry & Re-order Dispatch Engine

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall enforce FEFO picking order by prioritizing medicine batches with the nearest expiry date during dispensing.	High	Pass: The oldest-expiring valid batch is selected first. Fail: A newer batch is selected while an older valid batch exists.	Reduces medicine wastage and ensures proper stock rotation.
FR-002	Functional	The system shall allow the Pharmacy Clerk to record medicine batch details including medicine name, batch ID, quantity, and expiry date.	High	Pass: A batch is stored when all mandatory fields contain valid values. Fail: The system rejects incomplete or invalid data.	Accurate batch data is required for expiry tracking and FEFO dispensing.
FR-003	Functional	The system shall generate a FEFO dispensing list for a requested medicine using available non-expired batches.	High	Pass: The list contains only available, non-expired batches ordered by nearest expiry date. Fail: An expired or incorrectly ordered batch appears.	Helps staff dispense medicines in the correct order.
FR-004	Functional	The system shall monitor medicine inventory levels and identify medicines whose stock falls below their configured reorder threshold.	High	Pass: Any medicine with stock below its configured threshold is flagged for reorder. Fail: A below-threshold medicine is not flagged.	Prevents stock shortages.
FR-005	Functional	The system shall automatically generate and dispatch a purchase order to the Inventory Supplier when stock falls below the configured reorder threshold.	High	Pass: A purchase order containing the required medicine and reorder quantity is generated and dispatched when the threshold is crossed. Fail: No order is dispatched when the threshold condition is met.	Automates replenishment and reduces manual work.
NFR-001	Non-Functional – Performance & Security	The system shall generate and dispatch supplier purchase-	High	Pass: Benchmark testing confirms the defined latency target	Ensures timely and secure replenishment.

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		order requests when item stock falls below the configured threshold with low response latency and secure request handling.		and authorized secure request handling. Fail: Performance or security benchmark fails.	
NFR-002	Non-Functional – Reliability & Data Integrity	The system shall maintain accurate inventory records and prevent expired medicine batches from being selected for dispensing.	High	Pass: Testing confirms expired batches are excluded and inventory updates remain consistent. Fail: An expired batch is selected or inventory data becomes inconsistent.	Prevents unsafe dispensing and maintains reliable data.